

Cairo University

Faculty of Computers and Artificial Intelligence

Introduction to Database - IS211/SIS211

Project Phase 3 - Physical Data Model & SQL

Project Title: The Professional Beauty & Wellness Network

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Project Description

A luxury wellness brand manages several high-end spas that offer a variety of relaxation and grooming treatments. Each spa location features unique amenities and a specific number of private rooms. The brand employs certified therapists, each of whom possesses specific skills and is assigned to a primary spa location. Clients must create a membership profile to access the services. When a client wants a treatment, they book a session for a specific service (like a massage or facial) with a chosen therapist at a specific time. Each service has a set duration and a list of specialized organic products used during the session. The system must also track the consumption of these products, recording which items were used in which session and the quantity consumed to ensure the inventory is replenished automatically by the management.

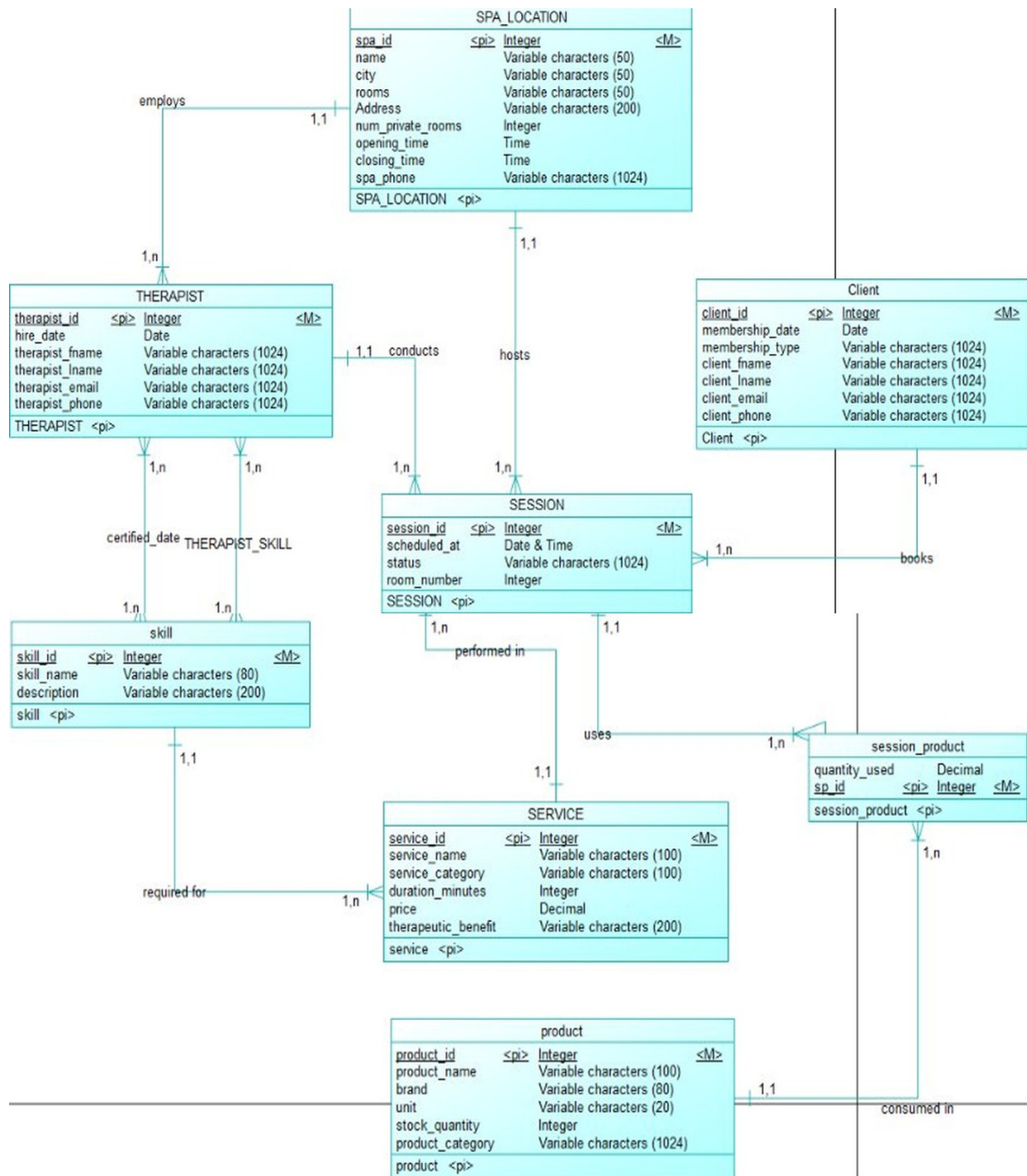
System Functionality

- Managing spa locations and the availability of private rooms.
- Registering therapists and mapping their specific treatment skills.
- Scheduling client sessions and managing therapist calendars.
- Tracking the usage of wellness products during specific treatments.
- Filtering services by duration, price, or therapeutic benefit.

System Inquiries

- Which treatment service was the most popular, having the maximum number of bookings?
- Which therapist had no client sessions scheduled during the last month?
- Who was the client who spent the most on premium category services last month?
- Identify services that were not booked by any client last month.
- What were the available therapists at each spa location last month?
- For each therapist, retrieve their profile and the total number of sessions they successfully completed.

Conceptual Entity Relationship Diagram (CDM)



Conceptual Data Model — The Professional Beauty & Wellness Network

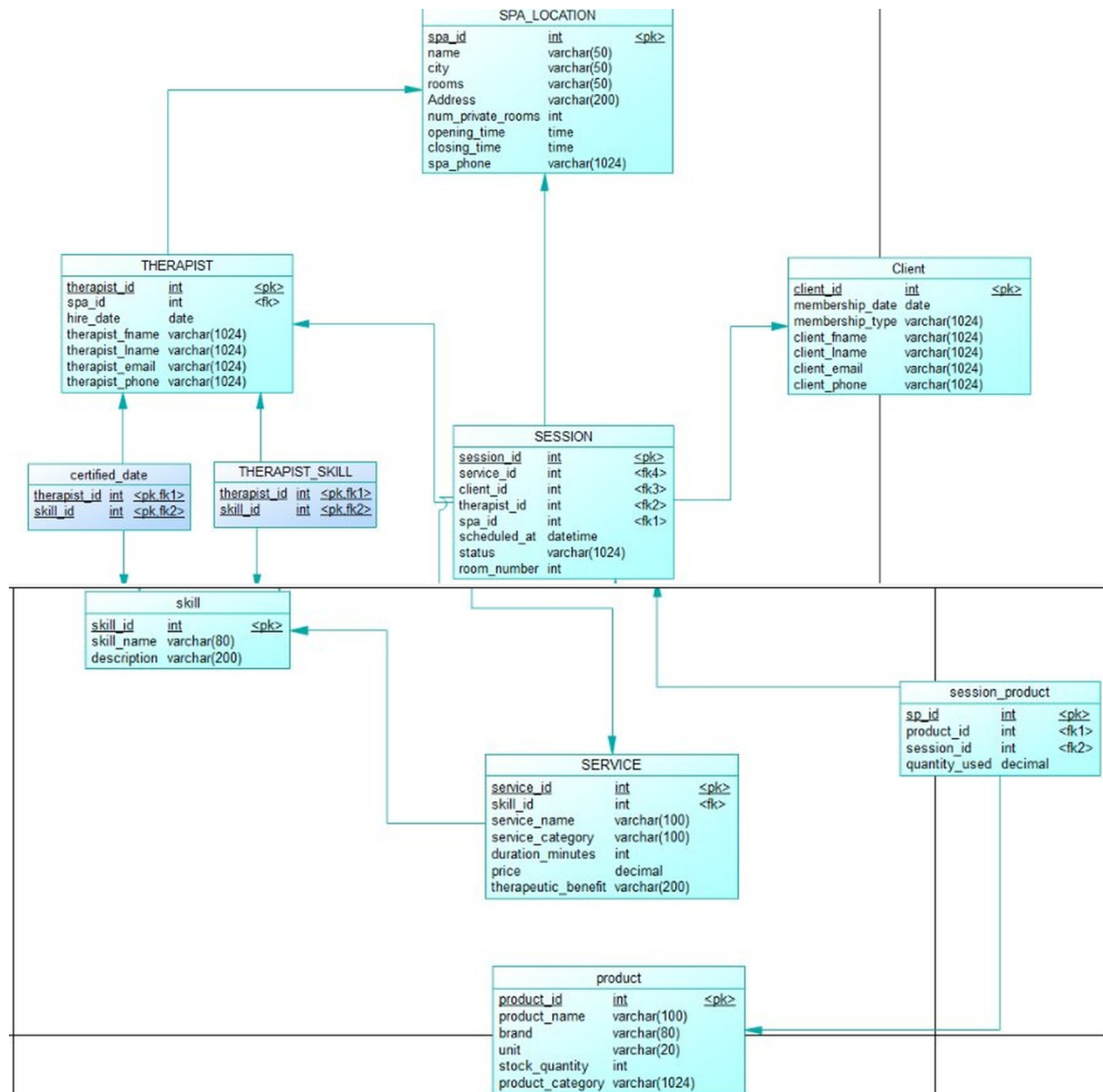
Entities & Attributes Summary

Entity	Key Attributes
SPA_LOCATION	spa_id (PK), name, city, rooms, Address, num_private_rooms, opening_time, closing_time, spa_phone
THERAPIST	therapist_id (PK), hire_date, therapist_fname, therapist_lname, therapist_email, therapist_phone
SKILL	skill_id (PK), skill_name, description
THERAPIST_SKILL	certified_date — Resolves M:N between THERAPIST and SKILL
CLIENT	client_id (PK), membership_date, membership_type, client_fname, client_lname, client_email, client_phone
SERVICE	service_id (PK), service_name, service_category, duration_minutes, price, therapeutic_benefit
SESSION	session_id (PK), scheduled_at, status, room_number
PRODUCT	product_id (PK), product_name, brand, unit, stock_quantity, product_category
SESSION_PRODUCT	sp_id (PK), quantity_used — Resolves M:N between SESSION and PRODUCT

Relationships Summary

From	Relationship	To	Cardinality
SPA_LOCATION	employs	THERAPIST	(1,1) — (1,n)
SPA_LOCATION	hosts	SESSION	(1,1) — (1,n)
THERAPIST	conducts	SESSION	(1,1) — (1,n)
CLIENT	books	SESSION	(1,1) — (1,n)
SERVICE	performed in	SESSION	(1,1) — (1,n)
SKILL	required for	SERVICE	(1,1) — (1,n)
SESSION	uses	SESSION_PRODUCT	(1,1) — (1,n)
PRODUCT	consumed in	SESSION_PRODUCT	(1,1) — (1,n)
THERAPIST	has skill	SKILL	(1,n) — (1,n)

Physical Entity Relationship Diagram (PDM)



Physical Data Model — The Professional Beauty & Wellness Network